

Minwoo Kim

Kookmin University, Seoul, Republic of Korea

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[Github](#) · [LinkedIn](#)

EDUCATION

Kookmin University, Seoul, Republic of Korea

Mar 2021 – Feb 2027 (Expected)

Bachelor of Science in Computer Science

- GPA: 3.97 / 4.50

RESEARCH INTERESTS

Autonomous and Cooperative Driving, Multi-Agent Perception, 3D Perception and Tracking, Vision-Language Models, Multimodal Learning, Robotics

RESEARCH EXPERIENCE

Dutt Research Group, University of California, Irvine

Apr 2026 – Aug 2026

Visiting Undergraduate Researcher | Advisor: Prof. Nikil Dutt

- Developing communication-efficient vision-language perception systems for resource-constrained edge intelligence
- Selected for the KMU–UCI Summer GREAT Program for collaborative research at UC Irvine

HCI Lab, Kookmin University

Jun 2025 – Present

Undergraduate Researcher | Advisor: Prof. Sangmin Yoon

- Developing video perception methods for 2D/3D object tracking and multimodal learning
- Research further supported by the SW & AI Research Program (UROP) at Kookmin University across three consecutive cohorts: Fall 2025, Winter 2025–26, Spring 2026

PUBLICATIONS

Rajat Bhattacharjya, **Minwoo Kim**, Nikil Dutt. *FreshLatent: Channel-Aware Latent Adaptation for Mission-Critical Embodied VLM Perception*. Under review, ASP-DAC 2027.

Minwoo Kim, Sangmin Yoon. *ContrastCoop: Cross-Agent Contrastive Alignment for Sparse Query Cooperative Perception*. Under review, AAAI 2027.

Minwoo Kim, Sangmin Yoon. *Memory as Dynamics: Learning Reliability-Guided Predictive Models for Online Video Perception*. ICML 2026.

Minwoo Kim, Sangmin Yoon. *SPARTA: Tracking by Selectivity with Policy-Gated Masks and Cross-Temporal Alignment*. Under review, IJCV.

HONORS & AWARDS

Award / Gold Award, Capstone Design Competition 2026

Honors / SW-Centered University Cohort Class Helper Fellowship 2026

Honors / SW-Centered University Research Excellence Fellowship 2025

Honors / SW-Centered University Academic Merit Scholarship 2025

Honors / Winter Research Laboratory Internship Fellowship 2025

Honors / SW-Centered University Cohort Class Helper Fellowship 2025

TALKS & PRESENTATIONS

Memory as Dynamics: Learning Reliability-Guided Predictive Models for Online Video Perception, ICML 2026, July 2026.

PERSONAL PROJECTS

Quantum Annealing-Based Optimization for CNN Tracking Models *Feb 2026 – Jun 2026*

- Formulated CNN channel pruning as a QUBO problem for quantum annealing-based optimization in visual tracking

Recipe Recommendation Service Based on User Preferences *Mar 2025 – Jun 2025*

- Developed a personalized recipe recommendation service with API-based metadata processing

TEACHING & LEADERSHIP

AI Programming Cohort Support Program, Kookmin University *Mar 2026 – Present*
Class Helper, AI Basic Programming

Undergraduate Research Seminar Program, Kookmin University *Sep 2025 – Present*
Seminar Leader, Computer Vision & Deep Learning

Computer Programming Cohort Support Program, Kookmin University *Mar 2025 – Jun 2025*
Class Helper, Computer Programming

SKILLS

Programming: Python, C/C++

ML / DL Frameworks: PyTorch

Languages: Korean (native), English